Zherui XIE

Tel: +1(669)-319-959 | Email: zheruix@andrew.cmu.edu
Carnegie Mellon University, Silicon Valley | Add: Sunnyvale, CA

EDUCATION

Carnegie Mellon University (CMU) 01/2023-Now
- Master of Science in Software Engineering

University of California, Davis (UC Davis) 08/2018-09/2021
- Bachelor of Science in Computer Science Overall GPA: 3.61/4.0; Upper-Division GPA: 3.85/4.0

PAPER

Zherui Xie, Masaaki Tsuzaki, Huimin Lu, Seiichi Serikawa, *Improved Point-Voxel Region Convolutional Neural Network for Small Object Detection*, in the 7th International Symposium on Artificial Intelligence and Robotics Conference (ISAIR), October 21-23, 2022; accepted by Society of Photographic Instrumentation Engineers (SPIE)

RESEARCH EXPERIENCE

Research Assistant(remote), Massachusetts Institute of Technology 08/2022-11/2022
Project: COVID-19 chest X-ray recognition algorithm based on convolutional neural network (CNN)
Advisor: Dr. Mark Vogelsbergerat
- Collected and organized open-source COVID-19 patient chest x-ray data;
- Optimized the CNN algorithm and train model on the dataset;
- Have been working on a paper

Lab Assistant(remote), Environment Recognition &Intelligent Computation Lab, Kyushu Institute of Technology 05-08/2022
Project: Algorithm Optimization of 3D Image Recognition
Advisor: Dr. Huimin Lu
- Optimized the 3D point cloud data processing algorithm based on PV-RCNN algorithm;
- Wrote a paper with other authors, which has been accepted by ISAIR 2022, and will be published by SPIE.

Research Assistant(remote), the Institute for AI Industry Research (AIR), Tsinghua University 03/2022-06/2022
Project: Pure Vision-Based UAV Navigation in Neural Radiation Fields
Advisor: Dr. Yongliang Shi
- Based on NeRF (Neural Radiance Fields), specifically the Mega-NeRF algorithm, rendered the images into 3D models and outputted them as .ply meshed map files;
- Creatively combined traditional PLY mesh maps with SDF (Signed Distance Field) files;
- Took charge of UAV route planning on the Nerf rendered map and used the Informed-RRT* algorithm for UAV path planning;
- Used the Minimum-jerk algorithm for the path planning to achieve a smoother path forward.

Research Assistant, Sichuan Energy Internet Research Institute Tsinghua University, Chengdu 01/2022-02/2022
- Developed an electricity and grid internet project based on genetic algorithm;
- Developed an algorithm for anomaly identification in grid based on machine learning;
- Developed a visualization algorithm for energy topology diagram of power system based on Python;
- Developed a human-computer interface for new energy simulation model based on Qt and C++.

Research Assistant, Sichuan Institute of Computer Sciences, Chengdu 07/2021-09/2021
Project: Table Data Auto-processing System Based on Convolutional Neural Networks (CNN) and Optical Character Recognition
Advisor: Professor Dezhong Peng
- Produced a smart form slicing system based on OpenCV's optical character recognition package, cut the form into small cells in order by processing the images, numbered and output each cell;
- Processed and labeled these cells manually and combined them into a dataset;
- Performed CNN model training on the dataset using Tensorflow and Python, trained the model to identify checkmarks in the questionnaire;
- Packaged all modules and integrated them into a Python-based data processing program for automatic counting and final machine learning results.

PROFESSIONAL EXPERIENCE

China Construction Bank Fintech 10/2021-12/2021
Algorithm Engineer (Intern), Center for Basic Technology Research Chengdu, China

- Learned mainstream algorithms and models of feature detection and target recognition, such as Yolo, CNN, RCNN, VGG, Fots, transformer, and LSTM, and some of the more classical algorithms such as Selective Search, Brief, Surf, and CTPN;
- Trained the Yolo and RCNN target detection model using the coco128 and image net dataset;
- Studied the source code of the CTPN algorithm, analyzed the structure of the algorithm, and used the algorithm to train a text-recognizable image.

Gamamobi Company 04/2021-06/2021
Program Developer (Intern), Game Development Department Chengdu, China

- Participated in the back-end game development with Lua, including the character auto path-search system based on A* search algorithm, character backpack management system and inventory management system;
- Realized the interface between server-side and customer-side using protobuf and Golang.

West China Medical College of Sichuan University 07/2020-09/2020
HIS (Hospital Information System) operator and maintainer, Hospital Information Technology Department Chengdu, China

- Participated in developing and optimizing new functions of the stored procedure of HIS and CIS (clinical information system) using SQL Server, and the optimization of some data transfer contents and structure;
- Conducted data maintenance of SQL Server database, including exception handling of user feedback during clinical use.

SELECTED COURSE PROJECTS

User-level Multithreading Library based on C	01/2021-02/2021
A Shell-like Command Line System Development based on C	01/2021
Oska Board Game Solver Compatible with Any Number of Board Layers Based on Minimax Algorithm	05/2020
Machine Learning for Human Activity Recognition Using the WISDM Dataset	06/2021
FAT-based Filesystem Software Stack based on C	03/2021
Mobile Operator Customer Churn Model Based on ANN and Python	11/2020
Linear Regression Library and LR Car Weight Model based on Python	10/2020
OpenStreetMap (OSM) Map File Processing based on C++	06/2019

TEACHING ASSISTANT

Computer Science Tutor, University of California, Davis 01/2021-03/2021

- Assisted lower-class students in algorithms, C++, C, Python and web programming.

ACADEMIC CONTEST

First Tech Challenge World Championship 03/2016-05/2016

- Programmed the robot using Java using a combination of McNamee wheels, infrared sensors, and gyroscopes to achieve precise automatic and manual control; led the team to second place in the preliminary round in Beijing and qualified for the final round in St. Louis.

ADDITIONAL INFORMATION

- **Programming:** C(advanced), C++(advanced) , Python(advanced), Java(advanced), Golang(intermediate), Erlang(intermediate), Haskell(intermediate), Prolog(intermediate), JavaScript(intermediate) , Lua(beginner)
- **Development Kit**: PyTorch(advanced), Tensorflow(advanced), Pandas(advanced), Seaborn (intermediate), Sklearn(intermediate)
- **Coursera Course**: Machine Learning, an online course from Stanford University
- **GRE:** V 158 (79%) + Q 170 (96%) + AW 3.5 (37%)